LABORATORY OF AUTOMATION SYSTEMS

Davide Samorì

Dipartimento di Ingegneria dell’Energia Elettrica e dell’Informazione (DEI)

Università di Bologna

Email: davide.samori@unibo.it

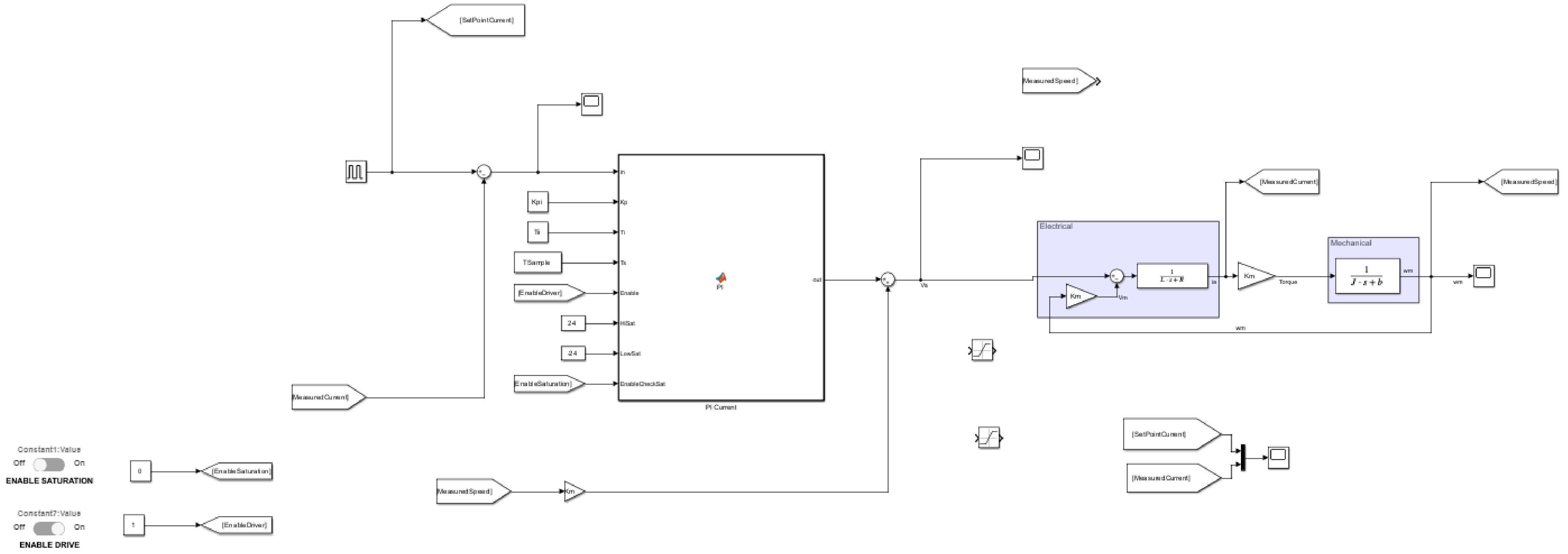
You can download and open:

- BaseModelForControl.slx
- BaseModelForControlParameters.m
- Your PI with DC Motor Model

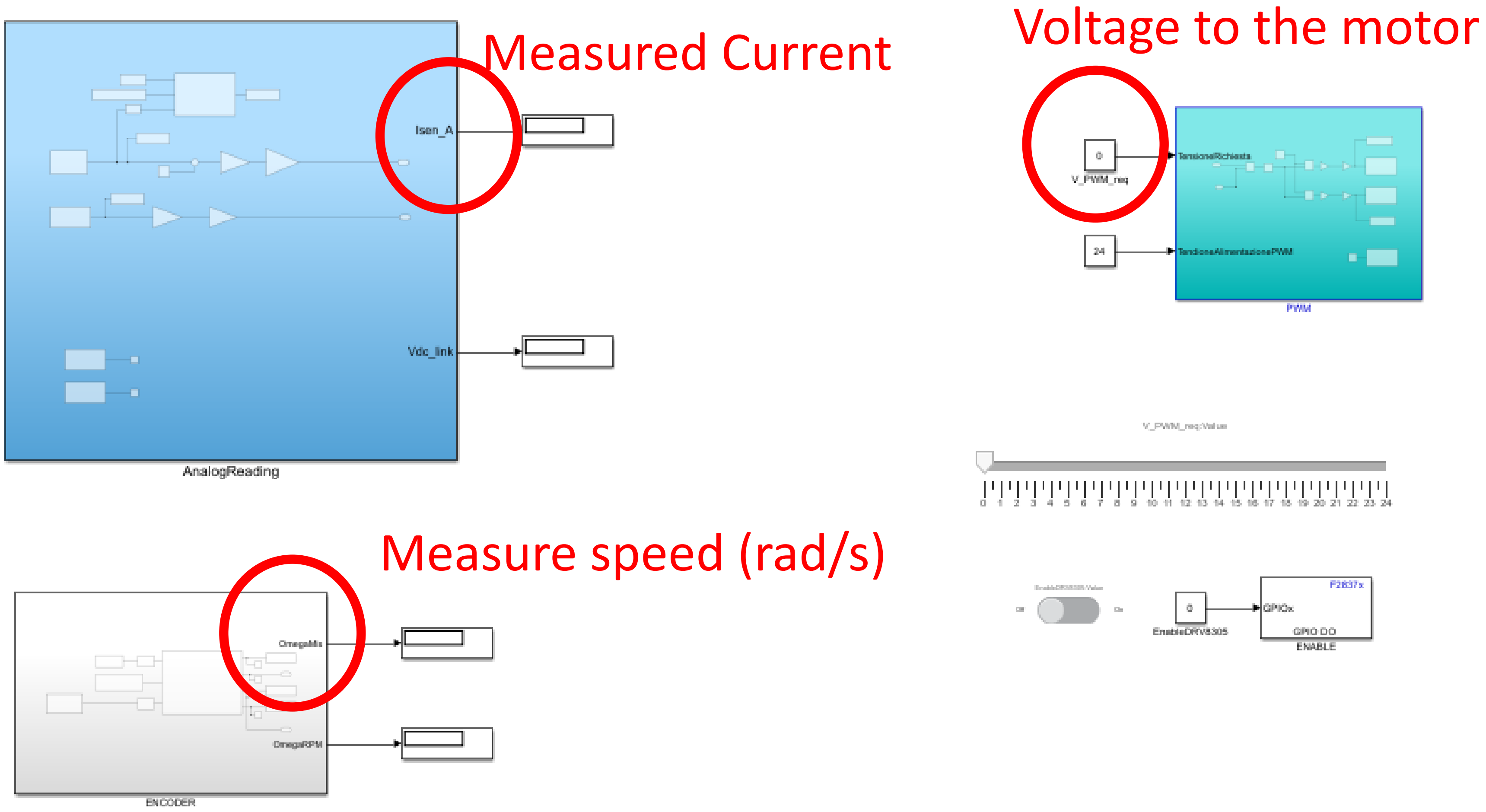
Today Goals:

- Run PI with motor

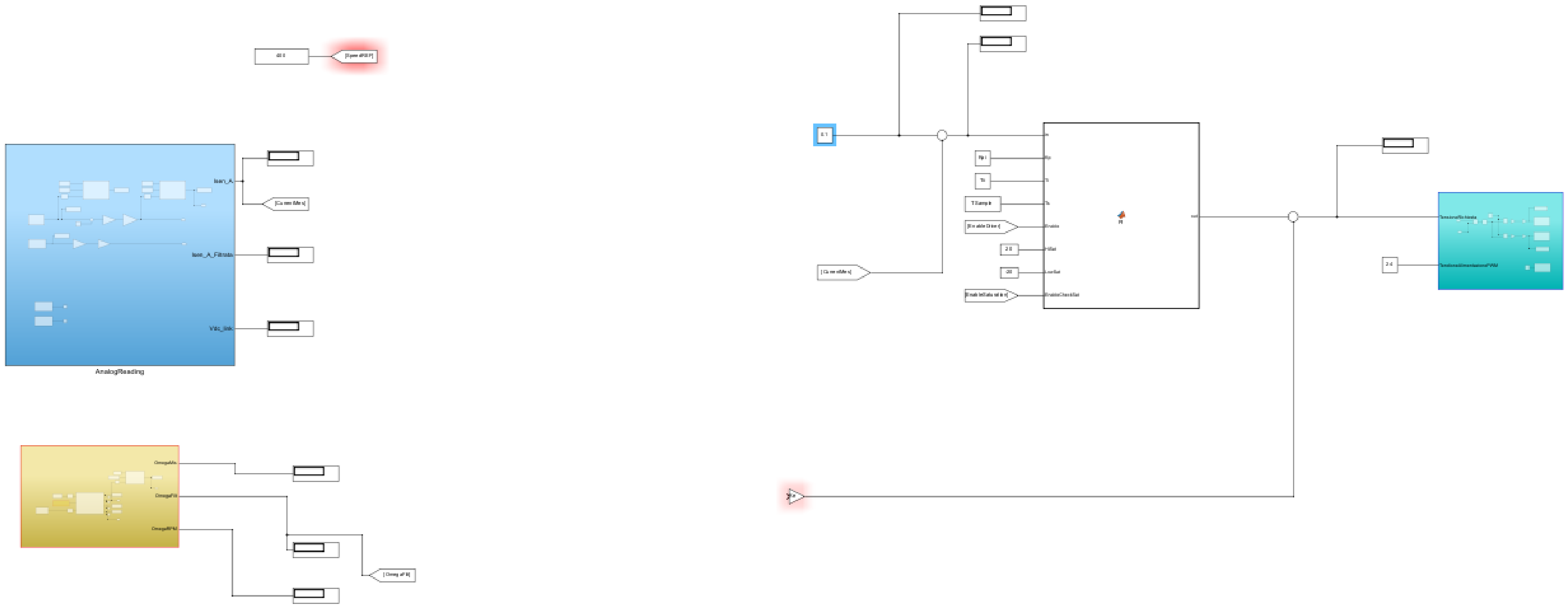
Last Time we tested PI (Current/Speed) on our DC Motor Model



You can copy your PI **in BaseModelForControl.slx**



Start with only current Loop



- **Add a display** on PI Output
- Add a display on error signal

- Connect Brake Resistor

- **Disable Saturation/AntiwindUp**

- Start with a SetPoint of 0.1A
- Increase with a step of 0.1 (0.2, 0.3, 0.4, ….) until the motor runs.
- Check error signal and other quantities

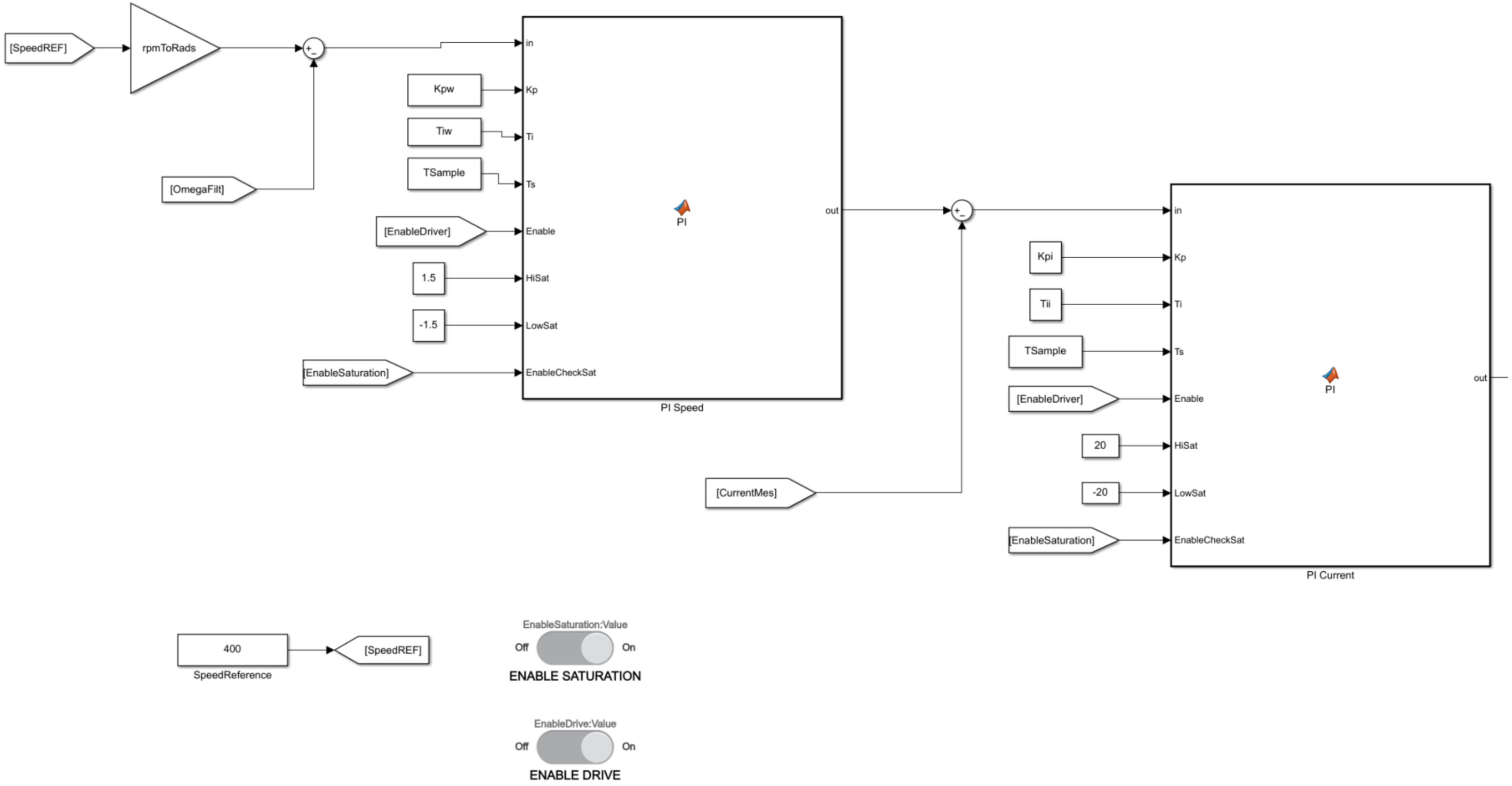
DISCONNECT BRAKE RESISTOR

- Start with a SetPoint of 0.1A
- Increase with a step of 0.1 (0.2, 0.3, 0.4, ….) until the motor runs.
- Apply a SetPoint of 0
- Apply a Set point of 1A
- Wait some seconds
- Apply a SetPoint of 0

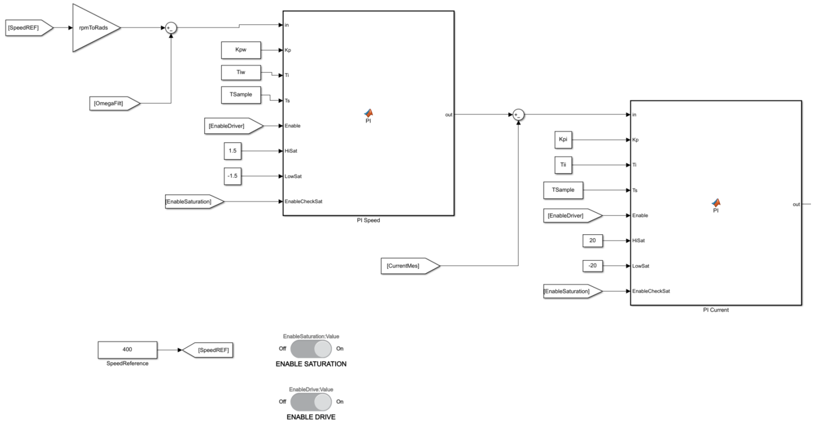
- **Did the engine stop?**
- **Why?**
- Watch what's happen to voltage

- **Do this steps again with AntiWindUp**

Now we can close also the speed loop



- pay attention to the unit of measurement of omega
- Try with or without load (watch voltage,…)
- What's about saturation value for Speed Loop?
    - Check if everything is OK



If everything's in order, we can take the next step….